

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

Looking at Relationships

Chapter 2 — Scatter Plots, Correlation, and Prediction

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Chapter Outline | What we will cover

Visualizing 2D Data
The Correlation Coefficient
Predicting from Correlation
Correlation Is Not Causation

Case Study: Wild Horse Herds
Pitfalls & Alternatives
Summary

Learning Objectives

- ▶ Visualize relationships with **scatter plots**, contingency tables, and heat maps
- ▶ Compute and interpret the **correlation coefficient** r
- ▶ Build the **best linear predictor** from r and estimate its error
- ▶ Recognize **spurious correlation** and confounding variables



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Section 1

Visualizing 2D Data

Categorical Data: Contingency Tables | Cross-tabulating two categorical variables

	0–1 goals	2+ goals
Male	18	27
Female	22	19

Reading a contingency table

Each cell is a joint count. Compare **row** or **column** proportions to spot association between gender and goals scored.

Avoid pie charts & 3D bar charts

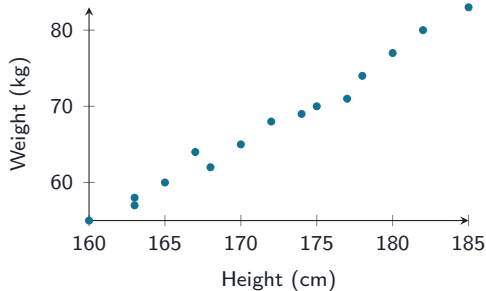
They distort area/angle perception and make comparison hard — prefer stacked bars or heat maps.

Scatter Plots | The default view for two numeric variables

How to read one

Each point is one data item (x_i, y_i) .

- ▶ Upward trend $\Rightarrow$ positive relationship
- ▶ Downward trend $\Rightarrow$ negative relationship
- ▶ Cloud with no trend $\Rightarrow$ no linear relationship



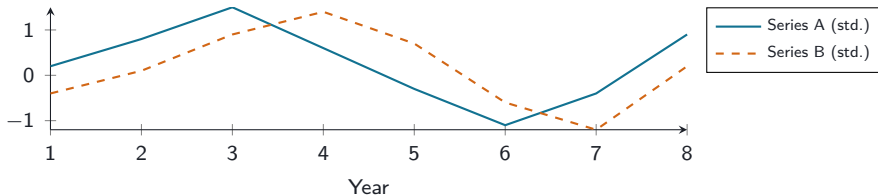
Time Series & Standardizing | Order matters; scale can mislead

Time series

Values indexed by time (burglaries per month, lynx pelts traded per year). Plot with a connected line to expose trend and seasonality.

Why standardize first?

Two series on different scales (e.g. dollars vs. counts) look incomparable until converted to **standard coordinates** $\hat{x} = (x - \bar{x})/\sigma$.



Section 2

The Correlation Coefficient

Defining Correlation | Quantifying a linear relationship

Definition

For standardized data $\{(\hat{x}_i, \hat{y}_i)\}$:

$$r = \text{corr}(\{(x, y)\}) = \frac{1}{N} \sum_{i=1}^N \hat{x}_i \hat{y}_i$$

Properties

- ▶ $-1 \leq r \leq 1$
- ▶ Symmetric: $\text{corr}(x, y) = \text{corr}(y, x)$
- ▶ Unaffected by translation or positive scaling

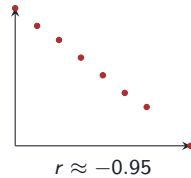
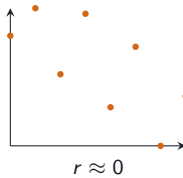
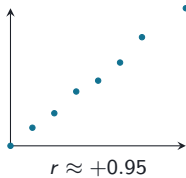
Interpreting the Sign and Magnitude | Direction, then strength

Value of r	Interpretation
$r = +1$	Perfect positive relationship ($\hat{y} = \hat{x}$)
$ r \geq 0.7$	Strong correlation
$0.3 \leq r < 0.7$	Moderate correlation
$ r < 0.3$	Weak correlation
$r = 0$	No <i>linear</i> relationship
$r = -1$	Perfect negative relationship ($\hat{y} = -\hat{x}$)

These are guidelines

Interpretation always depends on the domain — $r = 0.3$ can be remarkable in social science, unremarkable in physics.

Correlation Patterns | What different values of r look like



Section 3

Predicting from Correlation

The Best Linear Predictor | Deriving the least-squares line in standard coordinates

Predict $\hat{y}_i^p = a\hat{x}_i + b$. Requiring $\text{mean}(u) = 0$ for error $u = \hat{y} - \hat{y}^p$ forces $b = 0$. Minimizing $\text{var}(u) = 1 - 2ar + a^2$ over a gives:

$$\frac{d}{da}(1 - 2ar + a^2) = -2r + 2a = 0 \Rightarrow a = r$$

Result

In standard coordinates:

$$\hat{y}^p = r \hat{x}$$

In original coordinates:

$$y^p = \bar{y} + r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

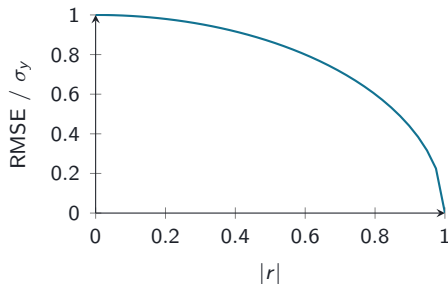
How Good Is the Prediction? | Error shrinks as $|r|$ grows

RMSE of the prediction

$$\text{RMSE} = \sigma_y \sqrt{1 - r^2}$$

Rule of thumb

If x is k std. devs. from its mean, y is predicted to be $r \cdot k$ std. devs. from its mean.



Perfect correlation ($|r|=1$) $\Rightarrow$ zero error



Section 4

Correlation Is Not Causation

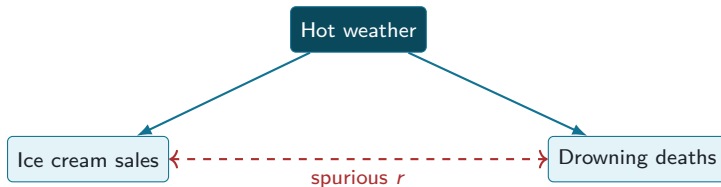
Spurious Correlation | Ice cream sales and drowning deaths

The naive observation

Ice cream sales and drowning deaths rise and fall together across the year, r is large and positive.

The real explanation

A **lurking variable** — hot summer weather — drives both *more swimming* and *more ice cream*.



The naive observation

Across all schoolchildren, shoe size correlates strongly with reading test scores.

The confound: age

Older children have *both* bigger feet and better reading skills. Age is the true driver of both.

Three explanations for any observed r

1. X causes Y
2. Y causes X
3. A hidden variable Z causes both X and Y

Section 5

Case Study: Wild Horse Herds

Case Study — Sterile Males in Wild Horse Herds | A correlation that reversed sign

The problem

Does introducing sterile males reduce foal births? A naive correlation between % sterile males and births per year was computed *across all herds and years*.

Surprising result

The naive correlation came out **positive** — more sterile males, *more* births?

The key insight: time as a lurking variable

Herd size and the sterile-male program both grew over **time**. Correlating *within* each year (controlling for time) reveals the expected **negative** relationship.

Section 6

Pitfalls & Alternatives

Three Pitfalls of Correlation | Where r can badly mislead you

Pitfall	What goes wrong
Non-linear relationships	$r \approx 0$ even for a strong curved pattern
Outliers	A single point can swing r dramatically
Scaling / grouping	Combining subgroups can flip the sign (Simpson-like effect)

Golden rule

Always plot the scatter plot before trusting a single number.

Spearman's Rank Correlation | A robust alternative to Pearson's r

Idea

Replace each value by its **rank**, then compute Pearson correlation on the ranks.

Why it's robust

- ▶ Detects any **monotonic** relationship, not just linear
- ▶ Outliers only shift a rank by one place

Correlation Matrix & Heat Maps | Scaling to many variables

	Height	Weight	Age	Income
Height	1.0	0.86	0.10	0.04
Weight	0.86	1.0	0.14	0.20
Age	0.10	0.14	1.0	0.48
Income	0.04	0.20	0.48	1.0

Reading a heat map

Darker cells = stronger correlation (in magnitude). Diagonal is always 1.0 (a variable with itself).



Section 7

Summary

Situation	Tool
Two categorical variables	Contingency table, stacked bar, heat map
Two numeric variables, linear	Scatter plot + Pearson r
Two numeric variables, monotonic	Spearman's rank correlation
Value indexed by time	Time series plot

Chapter Summary | Key takeaways

Visualize

Scatter plots first, always — correlation is only a number.

Quantify & predict

r measures linear association; the best linear predictor is $y^p = \bar{y} + r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$.

Never forget

Correlation $\neq$ causation — always ask about **lurking** and **confounding** variables.

Looking ahead

Part II builds the **probability theory** needed to formalize these ideas.

Thank you!

Questions & Discussion

Next: Chapter 3 — Basic Ideas in Probability